

```
!pip install -q spacy
!python -m spacy download en_core_web_sm
import nltk
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
```

Collecting en-core-web-sm==3.8.0

Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
12.8/12.8 MB 79.4 MB/s eta 0:00:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

[nltk_data] Downloading package punkt to /root/nltk_data...

[nltk_data] Unzipping tokenizers/punkt.zip.

[nltk_data] Downloading package punkt_tab to /root/nltk_data...

[nltk_data] Unzipping tokenizers/punkt_tab.zip.

[nltk_data] Downloading package stopwords to /root/nltk_data...

[nltk_data] Unzipping corpora/stopwords.zip.

True

```
from google.colab import files
uploaded = files.upload() # select 2.1_text_data.txt, 2.2_tokenization_data.txt, 2.3_clean_data.txt
```

Choose files 3 files

2.1_text_data.txt(text/plain) - 795 bytes, last modified: 01/09/2026 - 100% done

2.2_tokenization_data.txt(text/plain) - 672 bytes, last modified: 01/09/2026 - 100% done

2.3_clean_data.txt(text/plain) - 672 bytes, last modified: 01/09/2026 - 100% done

Saving 2.1_text_data.txt to 2.1_text_data (1).txt

Saving 2.2_tokenization_data.txt to 2.2_tokenization_data.txt

Saving 2.3_clean_data.txt to 2.3_clean_data.txt

```
import string

with open("2.1_text_data.txt", "r") as f:
    text = f.read()

print("Original Text:\n", text)
print("\n" + "="*60 + "\n")

uppercase_count = sum(1 for ch in text if ch.isupper())
lower_text = text.lower()
```

```

print(f"1. Uppercase letters found: {uppercase_count}")
print(f"    Lowercased text:\n    {lower_text}\n")

punctuation_found = [ch for ch in text if ch in string.punctuation]
punctuation_types = set(punctuation_found)
no_punct_text = lower_text.translate(str.maketrans('', '', string.punctuation))
print(f"2. Punctuation types found: {len(punctuation_types)} -> {punctuation_type}")
print(f"    Total punctuation marks: {len(punctuation_found)}")
print(f"    Text after removing punctuation:\n    {no_punct_text}\n")

numbers_found = [ch for ch in no_punct_text if ch.isdigit()]
no_num_text = ''.join([ch for ch in no_punct_text if not ch.isdigit()])
print(f"3. Numeric digits found: {len(numbers_found)}")
print(f"    Text after removing numbers:\n    {no_num_text}\n")

original_len = len(no_num_text)
cleaned_text = ' '.join(no_num_text.split())
extra_whitespace_chars = original_len - len(cleaned_text)
print(f"4. Extra whitespace characters removed: {extra_whitespace_chars}")
print(f"    Text after removing extra whitespace:\n    {cleaned_text}\n")

print(f"5. Final Cleaned Text:\n    {cleaned_text}")

```

Original Text:

Natural Language Processing (NLP) is a branch of Artificial Intelligence. It helps computers understand and process human language.

In 2026, NLP is used in many applications such as chatbots, machine translation, sentiment analysis, and text summarization!!!

Text preprocessing is an important step in NLP. It includes several operations such as cleaning text, removing punctuation, and removing unnecessary

For example, a chatbot may process 1000+ customer messages every day. These messages can contain CAPITAL letters, numbers like 12345, punctuation marks

The main goal of text preprocessing is to convert raw text into a clean and consistent format. This makes the text easier to analyze and use for furthe

=====

1. Uppercase letters found: 32

Lowercased text:

natural language processing (nlp) is a branch of artificial intelligence. it helps computers understand and process human language.

in 2026, nlp is used in many applications such as chatbots, machine translation, sentiment analysis, and text summarization!!!

text preprocessing is an important step in nlp. it includes several operations such as cleaning text, removing punctuation, and removing unnecessary

for example, a chatbot may process 1000+ customer messages every day. these messages can contain capital letters, numbers like 12345, punctuation marks

the main goal of text preprocessing is to convert raw text into a clean and consistent format. this makes the text easier to analyze and use for furthe

2. Punctuation types found: 6 -> {'!', '(', ')', '.', '+', ',', '}'

Total punctuation marks: 27

Text after removing punctuation:

natural language processing nlp is a branch of artificial intelligence it helps computers understand and process human language

in 2026 nlp is used in many applications such as chatbots machine translation sentiment analysis and text summarization

text preprocessing is an important step in nlp it includes several operations such as cleaning text removing punctuation and removing unnecessary num

for example a chatbot may process 1000 customer messages every day these messages can contain capital letters numbers like 12345 punctuation marks and

the main goal of text preprocessing is to convert raw text into a clean and consistent format this makes the text easier to analyze and use for further

3. Numeric digits found: 13

Text after removing numbers:

natural language processing nlp is a branch of artificial intelligence it helps computers understand and process human language

in nlp is used in many applications such as chatbots machine translation sentiment analysis and text summarization

text preprocessing is an important step in nlp it includes several operations such as cleaning text removing punctuation and removing unnecessary num

for example a chatbot may process customer messages every day these messages can contain capital letters numbers like punctuation marks and extra s

the main goal of text preprocessing is to convert raw text into a clean and consistent format this makes the text easier to analyze and use for further

4. Extra whitespace characters removed: 12

Text after removing extra whitespace:

natural language processing nlp is a branch of artificial intelligence it helps computers understand and process human language in nlp is used in ma

5. Final Cleaned Text:

natural language processing nlp is a branch of artificial intelligence it helps computers understand and process human language in nlp is used in ma

```
import re
import spacy
from nltk.tokenize import word_tokenize, sent_tokenize
```

```
with open("2.2_tokenization_data.txt", "r") as f:
    text = f.read()
```

```
print("Original Text:\n", text)
print("\n" + "="*70)
```

```
# --- NLTK ---
print("\n--- Using NLTK ---")
print("Words:", word_tokenize(text))
print("Sentences:", sent_tokenize(text))
```

```
# --- spaCy ---
print("\n--- Using spaCy ---")
```

```

nlp = spacy.load("en_core_web_sm")
doc = nlp(text)
print("Words:", [token.text for token in doc])
print("Sentences:", [sent.text.strip() for sent in doc.sents])

# --- Manual (no libraries) ---
print("\n--- Without NLTK/spaCy ---")
manual_sentences = [s.strip() for s in re.split(r'(?<=[!?])\s+', text.strip()) if s]
manual_words = re.findall(r"\b\w+(?:'\w+)?\b", text)
print("Words:", manual_words)
print("Sentences:", manual_sentences)

```

Original Text:

Natural Language Processing (NLP) is a fascinating field of Artificial Intelligence. It enables computers to understand human language. NLP is used in

Text tokenization is one of the first steps in NLP. Sentence tokenization divides a paragraph into individual sentences. Word tokenization divides each

For example, a chatbot receives thousands of messages every day. The system processes these messages and identifies useful information. This informatio

=====

--- Using NLTK ---

Words: ['Natural', 'Language', 'Processing', '(', 'NLP', ')', 'is', 'a', 'fascinating', 'field', 'of', 'Artificial', 'Intelligence', '.', 'It', 'enable

Sentences: ['Natural Language Processing (NLP) is a fascinating field of Artificial Intelligence.', 'It enables computers to understand human language.

--- Using spaCy ---

Words: ['Natural', 'Language', 'Processing', '(', 'NLP', ')', 'is', 'a', 'fascinating', 'field', 'of', 'Artificial', 'Intelligence', '.', 'It', 'enable

Sentences: ['Natural Language Processing (NLP) is a fascinating field of Artificial Intelligence.', 'It enables computers to understand human language.

--- Without NLTK/spaCy ---

Words: ['Natural', 'Language', 'Processing', 'NLP', 'is', 'a', 'fascinating', 'field', 'of', 'Artificial', 'Intelligence', 'It', 'enables', 'computers'

Sentences: ['Natural Language Processing (NLP) is a fascinating field of Artificial Intelligence.', 'It enables computers to understand human language.

```

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords

with open("2.3_clean_data.txt", "r") as f:
    text = f.read()

print("Original Text:\n", text)
print("\n" + "="*60 + "\n")

original_tokens = word_tokenize(text)
stop_words = set(stopwords.words('english'))

filtered_tokens = [w for w in original_tokens if w.lower() not in stop_words]

```

```
rint("Original Tokens:", original_tokens)
rint("\nStop Words Found:", found_stop_words)
rint("\nFiltered Tokens:", filtered_tokens)
rint(f"\nTotal: {len(original_tokens)} | Removed: {len(found_stop_words)} | Remaini
```

Original Text:

Natural Language Processing is an important field of Artificial Intelligence.

It is used in many applications such as chatbots, machine translation, sentiment analysis, and text summarization.

The main purpose of NLP is to enable computers to understand and process human language.

A computer can analyze a large amount of text in a short period of time.

The removal of stop words is one of the common steps in text preprocessing.

Stop words are commonly occurring words such as the, is, a, an, in, of, and, to, for, on, with, from, by, this, that, and it.

Removing these words can reduce the size of the text and improve the efficiency of some NLP applications.

=====

Original Tokens: ['Natural', 'Language', 'Processing', 'is', 'an', 'important', 'field', 'of', 'Artificial', 'Intelligence', '.', 'It', 'is', 'used', ' '

Stop Words Found: ['is', 'an', 'of', 'It', 'is', 'in', 'such', 'as', 'and', 'The', 'of', 'is', 'to', 'to', 'and', 'A', 'can', 'a', 'of', 'in', 'a', 'of

Filtered Tokens: ['Natural', 'Language', 'Processing', 'important', 'field', 'Artificial', 'Intelligence', '.', 'used', 'many', 'applications', 'chatbot

Total: 138 | Removed: 57 | Remaining: 81